Where the Admissible Radius Binds: a Correction and a Measured Boundary in Online Conformal Prediction

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Abstract

A printed claim that the scorecaster breaks online conformal coverage meets its refutation by overlapping authors; this paper joins them and locates the edge of the set that refutation relies on. Conformal PID states long-run coverage for any bounded scorecaster; a corollary keeps it for predictable perturbations of the deployed threshold inside an admissible radius. The claim was acted on; the readout's placement and derivation are conceded by name. At the null scorecaster that condition is *tight*. A legal dead band on the *completed* threshold leaves that set at its radius, and past it miscoverage is 1.000000 against $\alpha = 0.1$: coverage fails outright, not merely its rate. The edge belongs to the saturator–scorecaster pair, not the readout; elsewhere the condition is sufficient only. Under the equally legal $\hat{q} \equiv +b/2$ the failing band covers; at $\hat{q} \equiv -b/2$ both it and partial adjustment at $w = 0.999$ miscover.

1 Introduction

A refereed paper states that a scorecaster breaks Conformal PID's coverage guarantee. A later one, by the same first and last authors, proves in the same notation that a bounded perturbation of the deployed threshold does not. The later cites the earlier, so this is ordinary self-correction, and the joining is what this paper adds.

An online conformal method [Angelopoulos and Bates, 2023] moves a threshold each round. Forecast stability names, scores and prices that movement [Tunc et al., 2013, Godahewa et al., 2025, Van Belle et al., 2023, Pritularga and Kourentzes, 2024, Genov et al., 2026, Van Belle et al., 2026]; forecast verification priced temporal structure at fixed level earlier, matching predictive marginals on real producers [Gneiting et al., 2007, Pinson and Girard, 2012, Worsnop et al., 2018]. Coverage and mean length under-describe a deployed interval [Min et al., 2026, Vaze, 2026]. None of it is claimed here.

The record carries a claim and its refutation; this paper joins them. Conformal PID [Angelopoulos et al., 2023] adds a *scorecaster* $\hat{q}_{t+1}$ to a saturating error integrator, asking only that it be predictable and lie in $[-b/2, b/2]$: "any function of the past" (Theorem 1). Such a readout is already quantified over. The claim reads "[s]ince using a scorecaster (D-part) in C-PID breaks the theoretical coverage guarantee" [Li and Rodríguez, 2025], stated and acted on in its baselines paragraph, verbatim on refereed printed page 18443. It is refuted twice, with proof. Li et al. [2025, Corollary 2] add any predictable d_t, $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ with $\mu_t \in [0, 1]$ predictable, to the deployed threshold; since $\hat{q}_t + d_t$ still lies in $[-b/2, b/2]$ it is itself a legal scorecaster, so the result "has exactly the CPID error-integration form" and $|E_T| = o(T)$. The same generic-slot argument carries AcMCP's long-run coverage [Wang and Hyndman, 2024]. Both identifiers and "scorecaster" were searched across arXiv full text, arXiv abstracts, OpenReview, and Semantic Scholar's six citers of Li and Rodríguez [2025]: none returns both the claim and the corollary. Hu et al. [2026] still call scorecaster choice "arbitrary" and lacking "principled guidance" while asserting valid coverage: model selection, not validity. **We claim neither the placement nor the derivation.**

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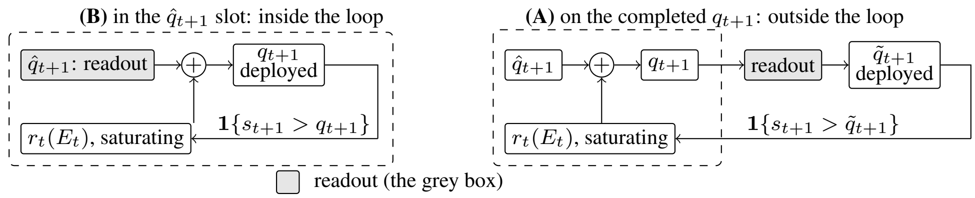

Figure 1: One scalar readout (grey) in its two places in (1); dashed is the loop r_t closes. In **(B)** the loop closes on the integrator's own output, and both readouts keep $\{\hat{q}_t\}$ in the convex hull of $\hat{q}_1$ and the raw scorecaster's range, inside Corollary 2's admissible set. In **(A)** it closes on $\tilde{q}_{t+1}$, which it never computed, and nothing keeps the deployed sequence inside that set.

Where Corollary 2 stops, its radius condition is tight. It is a *retention* result, silent outside that radius, and the question it leaves is whether the radius is sharp. Section 3 exhibits a legal perturbation leaving it: the L_1 dead-band family characterised here on the *completed* threshold, whose edge is the corollary's radius at the null scorecaster. Past it coverage fails outright, not merely the rate: the published sufficient condition is necessary there. Appendix C sets both claims beside four vocabularies for the same movement; Appendix D reads E_t as a capital process and builds a calibration test on it.

2 Setup

The producer. A forecaster fit before t induces a score s_t, and a threshold q_t is emitted before y_t is revealed. The deployed set is $C_t = \{y : s_t(x_t, y) \leq q_t\}$ and $\mathrm{err}_t = \mathbf{1}\{s_t(x_t, y_t) > q_t\}$. The target is long-run coverage, $T^{-1}\sum_{t\leq T}\mathrm{err}_t = \alpha + o(1)$, with no model on the data. Conformal PID [Angelopoulos et al., 2023] manipulates the threshold on the score scale, generalizing the running-error feedback that adaptive conformal inference [Gibbs and Candès, 2021] and its distribution-shift variant DtACI [Gibbs and Candès, 2024] already apply directly to the threshold:

$$q_{t+1} \;=\; \hat{q}_{t+1} \;+\; r_t\Big(\textstyle\sum_{i\leq t}(\mathrm{err}_i - \alpha)\Big), \tag{1}$$

Angelopoulos et al.'s iteration (5). Its r_t saturates: condition (4) requires $|r_t(x)| \geq b$ with the sign of x once $|x| \geq c\,h(t)$, for h nonnegative, nondecreasing and sublinear. With $E_t = \sum_{i\leq t}(\mathrm{err}_i - \alpha)$, Proposition 2 gives $|E_T| \leq c\,h(T) + 1$, read on realised scores.

One scalar, two readouts, two places. Damping a sequentially updated scalar has two standard readouts, both prior work. A quadratic cost gives linear partial adjustment $\tilde{q}_{t+1} = (1-\lambda)\hat{q}^{\mathrm{raw}}_{t+1} + \lambda\tilde{q}_t$ [Gârleanu and Pedersen, 2013], published as post-processing [Godahewa et al., 2025] and applied to a deployed threshold [Binny and Dixit, 2025, Eq. 13]. A proportional cost gives a no-trade band, moving only when the target is more than τ away, then pulling back exactly τ [Constantinides, 1986, Davis and Norman, 1990]. Either readout has two positions in (1), drawn in Figure 1; **(A)**, Placement A, is measured here. *The dead band (L_1) is not a stylistic alternative to the partial-adjustment (L_2) form*; it is the family whose failure Section 3 characterises. Neither is safe by construction: L_2 forfeits Proposition 2's rate while covering, L_1 past the radius loses coverage, and at the legal $\hat{q} \equiv -b/2$ both lose it. So the tight quantity is the radius, not either form.

The quantity, and the names it already has. We report the deployed threshold's *travel*, $\mathcal{T}_H = \sum_{t=2}^{H}|q_t - q_{t-1}|$, after actuator travel in process control. The arithmetic is not new. Over a predictive distribution $\sum|\Delta q|$ is the Multi-Quantile Change [Zanotti, 2026, v3, §3.4.1]; verification calls it *jumpiness*, a *convergence index*, *(in)consistency* [Zsótér et al., 2009, Ehret, 2010, Pappenberger et al., 2011]; applied control, IACER [Sarhadi, 2025]; dynamic-regret online learning, a path length. Appendix B names ten across four vocabularies, so a fresh one is used here.

Harness. The harness fixes $\alpha = 0.1$ and $b = 2$. Its saturator $r_t(x) = b\,\mathrm{clip}(x/(c\,h(t)), -1, 1)$, with $c = 1$ and $h(t) = \log(t+2)$, meets (4) with equality. The scorecaster is null and legal, reducing (1) to the pure error integration Proposition 2 addresses. Neither choice is neutral: (4) is a lower bound met exactly here, and Theorem 1 permits scorecasters $b/2$ either side of null, predictable but not constant in t. The adversary is specified, not tie-broken. It plays $s_t = \mathrm{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$, $\varepsilon = 10^{-9}$, against the *deployed* $\tilde{q}_t$: the $\varepsilon \downarrow 0$ reading of "sets the score at the threshold", firing the strict indicator. Legality is Theorem 1's $[-b/2, b/2]$ on both sides, so $b/2 + b/2 = b$ is exactly tight. *That specification names one member of a class*: any adapted $s_t \in [-b/2, b/2]$ is legal, the constant $s_t \equiv -b/2$ included. Every number in Table 2 (App. A) is measured against the one adversary above; Section 3's two boundaries are quantified over the class, and they differ.

3 The edge of the admissible set, located and proved

The retention result is published, and tested from outside. Li et al. [2025, Corollary 2] prove that a predictable d_t with $|d_t| \le \mu_t(b/2 - |\hat{q}_t|)$ on the deployed threshold keeps (1) in error-integration form and *retains* $|E_T| \le c\,h(T) + 1$: the radius is theirs. A downstream partial adjustment of weight w leaves it; Table 2 (App. A) prices that departure, and five of its eleven arms exceed the bound at $T = 10^6$, and for $w = 0.99$ and 0.999 the excursion is *flat in* T while the bound grows like $\log T$. The forfeit is a property of the smoothing *strength*, not Placement A, and it buys something: travel falls from 28,311 to 202. A readout on the completed q_{t+1} touches neither r_t nor the integrator's argument, so it cannot put (4) at risk. It delays a correction already computed, so the accumulator excurses *further*: smoothing the output does not damp E_t but feeds it.

Table 1: Placement A, primary regime, $T = 10^6$: four arms of Table 2 (App. A), which lists the remaining seven and finer detail. Proposition 2's bound is 14.8155; "ratio" is $\max_t |E_t|$ over it.

Arm	miscov. 10^6	$\max_t \lvert E_t \rvert$ 10^6	ratio
none (control)	0.100007	7.80	0.53
partial adj. $w = 0.999$	**0.100004**	623.70	42.10
dead band $\tau = 0.9$	0.100012	13.20	0.89
dead band $\tau = 1.5$	**1.000000**	900,000	60,747

The L_1 dead band leaves the radius, and the edge is a theorem. A dead band pulls the deployed threshold back by exactly τ when it releases and holds it otherwise, so $|\tilde{q}_{t+1} - q_{t+1}| \le \tau$ at every round and "inside Corollary 2" reads $\tau \le \mu_t(b/2 - |\hat{q}_t|)$. Under saturation the deployed value is pinned τ below the integrator's ceiling, and the deployed value's realised supremum is $b - \tau$ to the printed digit at every $\tau \ge 0.9$. Outside the radius two levels decide the answer and condition (4) names only one of them. Write $A_t^+ = \sup_x r_t(x)$ and $A_t^- = \inf_x r_t(x)$ for the integrator's extremes, and $\Lambda_t^+ = \inf_{x \ge c\,h(t)} r_t(x)$, $\Lambda_t^- = \sup_{x \le -c\,h(t)} r_t(x)$ for the levels it is *guaranteed* to reach once the accumulator clears the radius. Condition (4) says exactly $\Lambda_t^+ \ge b$ and $\Lambda_t^- \le -b$; that $\Lambda_t^\pm = A_t^\pm$, that the integrator reaches its extremes where (4) puts them, is a further hypothesis and not one (4) supplies. Set

$$\tau^{\star\pm} \;=\; \sup_t \big(|A_t^\pm| \pm \hat{q}_{t+1}\big) - b/2, \qquad \sigma^\pm \;=\; \inf_t \big(|\Lambda_t^\pm| \pm \hat{q}_{t+1}\big) - b/2. \tag{2}$$

Failure, over the whole admissible class. If $\tau > \tau^{\star+}$ then $\hat{q}_{t+1} + A_t^+ - \tau < b/2$ at every round, so $\tilde{q}_{t+1} \le \max(\tilde{q}_t, q_{t+1} - \tau)$ makes $\{\tilde{q} < b/2\}$ absorbing: a release upward lands below $b/2$, and a hold or a release downward cannot lift the deployed value at all. Against $s_t = \mathrm{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$ the indicator is then $\mathbf{1}\{\tilde{q}_t < b/2\}$ and every round misses. If $\tau \ge \tau^{\star-}$ the constant legal play $s_t \equiv -b/2$ makes $\{\tilde{q} \ge -b/2\}$ absorbing by the mirror inequality, so no round can miss and the realised rate is 0 rather than α. Either way long-run coverage is lost, with $|E_T|$ linear in T. Nothing here asks $\hat{q}$ to be constant, asks the extremes to be attained anywhere, or asks $A_t^\pm$ to be finite beyond what the hypothesis already forces. **Retention, keyed to the reach rather than the supremum.** If $\tau \le \sigma^+$ then $E_{t-1} \ge c\,h(t-1)$ forces $q_t \ge b/2 + \tau$ and so $\tilde{q}_t \ge b/2$, whence $\mathrm{err}_t = 0$ against *every* legal adversary; if $\tau < \sigma^-$ the mirror forces $\mathrm{err}_t = 1$ below $-c\,h(t-1)$. The accumulator cannot then leave $[-c\,h(t), c\,h(t)]$ by more than one round, and $|E_T| \le c\,h(T) + 1$: Proposition 2's own bound, retained verbatim rather than weakened to $o(T)$. Using condition (4) alone this already certifies $\tau \le b/2 + \inf_t \hat{q}_t$ and $\tau < b/2 - \sup_t \hat{q}_t$, which for a constant scorecaster is Corollary 2's radius at $\mu_t = 1$.

The two directions meet, with no gap, on the sub-class the five settings inhabit. If $\hat{q}_t \equiv \hat{q}$, if $\Lambda_t^\pm = A_t^\pm$, and if neither depends on t, then $\sigma^\pm = \tau^{\star\pm}$ and **coverage is retained against every legal adapted adversary if and only if** $\tau \le \tau^{\star+}$ **and** $\tau < \tau^{\star-}$**.** For a symmetric saturator that is one line: coverage if and only if $\tau \le \sup_x r_t(x) - b/2 - |\hat{q}|$, with the endpoint included only when $\hat{q} < 0$. Every setting run here is in that sub-class, so the sharp form applies to all five. **What was printed**

before is the failure half. $\tau^{\star+}$ is the rule Figures 2 and 3 (App. A) draw, and all 19 dead-band runs agree with it read against the adversary the harness plays. It is not a retention condition, and Table 3 (App. A) separates the two with legal objects only. The supremum can sit where the dynamics never go: $|E_t| \le (1-\alpha)t$, so a saturator equal to the harness's inside $|x| \le t^2$ and to $\pm 10b$ outside it satisfies (4), reads $\tau^{\star+} = 19$, and fails at $\tau = 1.001$. A legal time-varying $\hat{q}$ splits $\sup_t$ from $\inf_t$: $\hat{q}_t = b/2$ on t a power of two and 0 elsewhere reads $\tau^{\star+} = 2$ against $\sigma^+ = 1$. And that printed rule omits $\tau^{\star-}$ entirely.

The sharp form is one number, and its hypotheses are what buy it. Specialise to a symmetric saturator that meets (4) with equality and attains its extremes there ($\Lambda_t^{\pm} = A_t^{\pm} = \pm b$; the harness's $r_t = b\,\mathrm{clip}(x/(c\,h(t)), \pm 1)$ is one, Section 2), and to a constant $\hat{q}$. Then

$$\min(\tau^{\star+}, \tau^{\star-}) \;=\; b/2 - |\hat{q}|, \tag{3}$$

with the endpoint admissible only when $\hat{q} < 0$: exactly Corollary 2's radius at $\mu_t = 1$, now as an if-and-only-if. **The necessity direction above assumes none of it**: $\tau > \tau^{\star+}$, or $\tau \ge \tau^{\star-}$, forfeits coverage against every admissible r_t and every legal predictable $\{\hat{q}_t\}$, extremes attained or not, and that unrestricted statement is what this section claims. The two hypotheses are load-bearing rather than decorative: the unattained supremum and the varying-$\hat{q}$ split two paragraphs above are exactly what they rule out. Equation (3) is where the reach meets the supremum, not evidence that they always do.

The mirror boundary is already measured, on this paper's own grid. The τ-cliff run's exact-threshold column under the strict indicator is the mirror adversary $s_t \equiv -b/2$ (one error process, $\mathrm{err}_t = \mathbf{1}\{\tilde{q}_t < -b/2\}$), and on the same eleven widths at the null scorecaster it locates $\tau^{\star-} = 1$, closed on the failing side, exactly where (2) puts it; Appendix A prints the eleven readings. Both 1.000000 and 0.000000 are failures of the same order, $|E_T| = (1-\alpha)T$ and αT, **so the failure criterion is $|E_T|$, not the miscoverage rate**; and once $\hat{q}$ varies the rate stops being a tell at all.

Both degenerate edges, and what they need. An unbounded supremum is not on its own enough for $\tau^{\star+} = \infty$ to mean what it looks like: the same unreachable-set construction with $r_t(x) = x$ past $|x| > t^2$ has no finite supremum and fails at $\tau = 1.5$. What the tangent family has, and that construction does not, is a genuine reach: $\Lambda_t(X) = \tan(\arctan(b)X/(c\,h(t)))$, so the accumulator level at which the deployed threshold is forced to $b/2$ is $\kappa(\tau)\,c\,h(t)$ with $\kappa(\tau) = \arctan(b/2 + \tau - \hat{q})/\arctan(b)$, finite at every finite τ. Coverage is retained at every width, as measured; but $\kappa(\tau) > 1$ once $\tau > b/2 + \hat{q}$, so Proposition 2's own *constant* is forfeited past $\tau = b/2$ while coverage is not. At $\tau = 1.5$, $T = 10^6$ the excursion is 15.10 against Proposition 2's 14.8155 and against $\kappa(1.5)\,c\,h(T) + 1 = 15.8530$. At the other edge, $\sup_t \hat{q}_t = -b/2$ forces $\hat{q} \equiv -b/2$, since every $\hat{q}_t$ is at least $-b/2$ already. There $q_{t+1} - \tau \le -b/2 + b - \tau < b/2$ for every $\tau > 0$ and every t, and $\tilde{q}_1 = 0 < b/2$, so the failing set is absorbing from round 1 with no transient: the admissible window is the single point $\tau = 0$, and $\tau = 0.5$ returns 1.000000, as does partial adjustment at $w = 0.999$.

What that establishes is that the published condition is tight. At $\mu_t = 1$, $\hat{q} \equiv 0$ the boundary is exactly Corollary 2's radius, crossing it costs coverage rather than the rate ($\tau = 0.9$ covers at 0.100012, $\tau = 1.5$ returns 1.000000 with $\max_t |E_t| = 900{,}000$, 60,747 times Proposition 2's bound at $T = 10^6$; Table 2, App. A), and (2) makes that an if-and-only-if against the whole legal adversary class rather than a pattern across five settings. **The claim is that necessity: a tightness result about someone else's sufficient condition**, smaller and more precise than a forfeit of the rate by post-processing. Elsewhere it stays sufficient only, and now provably so: under the tangent integrator Corollary 2 permits $\tau \le b/2$ while no finite width forfeits coverage against any legal adversary, because the reach is genuine at every level. At $\hat{q} \equiv +b/2$ the looseness goes the other way and disappears: Corollary 2 permits radius 0, and (2) permits no width at all against the mirror adversary, the unsmoothed control included, though $\tau = 1.5$ covers there against the specified one. That asymmetry makes "tight" a located claim.

The constructive reading, and it is the corollary's rather than ours. Placement B keeps $\{\hat{q}_t\}$ inside the admissible radius, so Theorem 1 carries it already. What Section 3 adds is that the band has two edges, both computable before deployment from the integrator's guaranteed reach and the scorecaster's range, that crossing either is not a graceful degradation, and that between $\sigma^{\pm}$ and $\tau^{\star\pm}$, which separate only when $\hat{q}$ varies in t, neither direction is settled here.

4 Limitations

The retention direction is proved with no gap only where the two boundaries meet: a constant scorecaster and a saturator attaining its extremes at the condition-(4) radius, which is every setting run here; a genuinely time-varying $\hat{q}$ leaves the band $\sigma^+ < \tau \leq \tau^{\star+}$ open in neither direction (Section 3). And $0.63/(1-w)$ is specific to the saturator's *level*, which condition (4) bounds only from below. **The inherited guarantee is weak where it applies.** Under the tangent integrator $h(t) = t/\log t$, the certified gap $(c\,h(T)+1)/T$ decays as $O(1/\log T)$ rather than $O(1/T)$. What is forfeited was not a tight certificate. **The dead-band widths are absolute, and the failure is total only against the adversary.** τ is in score units with $b = 2$, so the failing $\tau = 1.5$ is $0.75\,b$. Under i.i.d. scores it returns 0.249376 at $T = 10^6$, an adversary-free confirmation of the mechanism rather than a retreat. The threshold pins at $b - \tau = 0.5$, and uniform scores on $[-b/2, b/2]$ put $P(s > 0.5) = 0.25$ exactly. **The harness was rebuilt against numbers already read**, which carries less weight than a rerun done without sight of them. **Placement B is conceded, not tested. The i.i.d. check above used one seed** (20260820, $T = 10^6$); **the primary, adversarial harness uses none**, and **only its null-scorecaster edge is bracketed**; the others are one-sided. **What remains open is the band between the two boundaries.** For a genuinely time-varying $\hat{q}$, coverage in $\sigma^+ < \tau \leq \tau^{\star+}$ is neither proved nor falsified; Table 3's power-of-two rows sit there, and closing that band is future work, not a sweep this paper has already run.

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Code and data

The simulator, its frozen configuration, the `results/` files behind every figure and table, the generators that read them, and this project's audit trail across sessions are at `https://[REPOSITORY LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION]`.

A Supporting measurements

The primary regime. Every number in Table 2 is measured under one deterministic adversary, which consumes no randomness and needs no seed. It plays $s_t = \mathrm{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$ with $\varepsilon = 10^{-9}$ against the *deployed* threshold, and the indicator is strict. The specification is not a tie rule, and the distinction is measured rather than argued. Scoring an exact-threshold play *as a miscoverage* reproduces every quantity Table 2 reports: 44 of 44 arm-horizon cells identical, all eleven arms at all four horizons. Scoring the same play under the *strict* indicator does not, and it moves every row, because the adversary's signature move then covers rather than misses. Partial adjustment at $w = 0.999$ excurses 70.80 in place of 623.70, and the failing dead band returns 0.000000 in place of 1.000000. The $\varepsilon \downarrow 0$ limit the harness specifies is the first reading, and that 623.70 turns on the adversary's ε, not the counting convention. **No row is convention-free with respect to the second**, which is why the adversary is specified rather than tie-broken. At $\tau = \tau^{\star+}$ itself the two readings part:

Table 2: Placement A: a one-scalar readout on the *completed* threshold, under one deterministic adversary, on one pass to $T = 10^6$; all eleven arms run are listed. A four-arm subset of this table – the control, the strongest partial-adjustment arm, and the two dead-band widths bracketing $\tau^{\star+}$ – appears in the body as Table 1; this table adds the remaining seven arms, the $T = 2{\times}10^5$ horizon, saturation fraction, and threshold travel the body table omits. Proposition 2's bound is 13.2061 at $T = 2{\times}10^5$ and 14.8155 at $T = 10^6$; "ratio" is $\max_t |E_t|$ over it, and "Sat." the fraction of rounds on which (4) delivers its full $\pm b$. Every number is the *primary* regime's, defined in this appendix, which also gives the $T = 10^4$ excursions and the fitted excursion law.

Readout on the completed threshold	miscov. $2{\times}10^5$	miscov. 10^6	$\max_t \lvert E_t\rvert$ 10^6	ratio	Sat. 10^6	travel 10^6
none (control)	0.100035	0.100007	7.80	0.53	0.0000	28,311
partial adj. $w = 0.5$	0.100030	0.100007	7.70	0.52	0.0000	18,113
partial adj. $w = 0.9$	0.100030	0.100007	7.50	0.51	0.0000	4,081
partial adj. $w = 0.99$	0.100040	0.100006	63.00	4.25	0.0007	1,023
partial adj. $w = 0.999$	0.100035	**0.100004**	623.70	42.10	0.0097	202
growing time constant $p = 0.5$	0.100030	0.100006	12.60	0.85	0.0000	330
growing time constant $p = 0.9$	0.100030	0.100024	53.00	3.58	0.3181	16
running mean	0.099940	0.100010	49.50	3.34	0.3647	9
dead band $\tau = 0.5$	0.100035	0.100005	10.60	0.72	0.0000	2,045
dead band $\tau = 0.9$	0.100060	0.100012	13.20	0.89	0.0038	1,051
dead band $\tau = 1.5$	**1.000000**	**1.000000**	900,000	60,747	1.0000	0.5

the strict indicator returns 0.000000 where scoring the exact play as a miscoverage returns 1.000000. **Neither of those is coverage.** They are $|E_T| = \alpha T$ and $(1-\alpha)T$, failures of the same order and opposite sign. Under the strict indicator the exact-threshold play scores $\mathrm{err}_t = \mathbf{1}\{\tilde{q}_t < -b/2\}$, which is also the indicator of the constant legal play $s_t \equiv -b/2$, so that column is Section 3's *mirror adversary* rather than a second convention. It returns between 0.09992 and 0.09997 at $\tau = 0.5$, 0.9, 0.95, 0.99 and 0.999, and 0.000000 with $\max_t |E_t| = 10{,}000$ at $\tau = 1$, 1.001, 1.01, 1.05, 1.1 and 1.5. That locates $\tau^{\star-} = 1$ on the same eleven widths Figure 2 uses, closed on the failing side. The convention decides which legal adversary realises the endpoint failure, not whether one occurs.

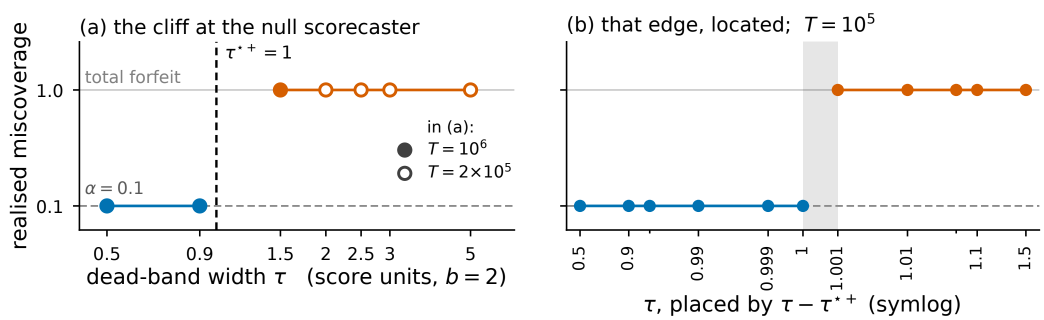

Figure 2: Realised miscoverage against dead-band width τ at the null scorecaster, $\hat{q} \equiv 0$ with a saturator of level b, under the *specified* adversary, where the upward boundary $\tau^{\star+}$ of (2) is 1. **(a)** The full sweep: filled markers run to $T = 10^6$ and the four widest bands, drawn open, to $T = 2{\times}10^5$; blue covers, orange forfeits, and the dashed rule is $\tau^{\star+}$. **(b)** The same setting at $T = 10^5$ over eleven measured widths, placed by $\tau - \tau^{\star+}$ on a symlog axis whose linear zone is $|\tau - \tau^{\star+}| \le 10^{-3}$, which magnifies the neighbourhood of $\tau^{\star+}$: visual spacing here is not proportional to true distance in τ. Unlabelled minor ticks mark $\tau = 0.95$ and 1.05. The grey band spans the widest covering width to the narrowest failing one; Figure 3 gives the other four $(r_t, \hat{q})$ settings. The same eleven widths under the *mirror* adversary, which locates $\tau^{\star-}$, are read off in this appendix's primary-regime paragraph.

Excursions at $T = 10^4$, and where the excursion law holds. Partial adjustment excurses 6.30, 63.00 and 623.70 at $w = 0.9$, 0.99 and 0.999. The last two are flat in T while Proposition 2's bound grows like $\log T$, which is why the ratio falls with the horizon: at $w = 0.999$ it is 61.08 at $T = 10^4$ and 42.10 at $T = 10^6$. On the control and the four partial-adjustment arms the excursion fits $\max_t |E_t| \approx \max\{0.5\, h(T) + 0.9,\ 0.63/(1-w)\}$ at every horizon, most loosely at $w = 0.9$, the one arm whose two branches cross inside the horizon range. This is a fit to those five arms, not a derived bound. It does not extend to the other six: the growing time constants, the running mean and the dead bands all excurse further than it predicts.

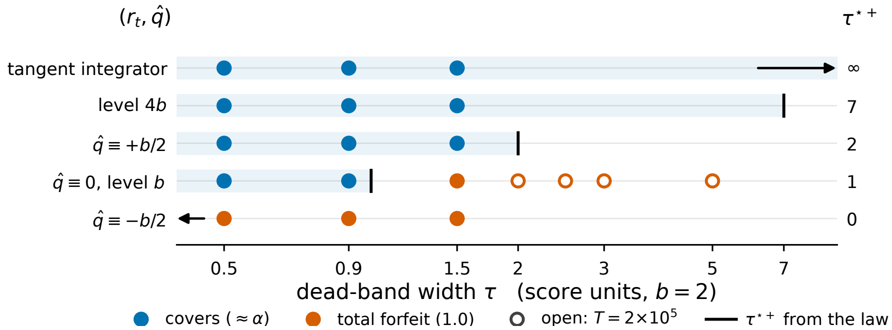

Figure 3: All 19 dead-band runs, at the five admissible $(r_t, \hat{q})$ settings: rows are settings ordered by $\tau^\star$, and the shared horizontal axis is τ. Blue markers cover at $\approx \alpha$ and orange markers return 1.000000; filled markers run to $T = 10^6$ and open markers to $T = 2{\times}10^5$, all under the *specified* adversary. The rule on each row is that setting's upward boundary $\tau^{\star+} = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$, printed at the right, and the tinted band is $\tau \leq \tau^{\star+}$. Two rules fall off the axis and are drawn as arrows: $\tau^{\star+} = 0$ at $\hat{q} \equiv -b/2$, and $\tau^{\star+}$ unbounded under Angelopoulos et al.'s tangent integrator. The mirror boundary $\tau^{\star-}$ of (2) is not drawn here and differs from $\tau^{\star+}$ at the two non-null scorecasters.

Realised miscoverage at the covering settings of Figure 3. At $\hat{q} \equiv +b/2$ the $\tau = 1.5$ band returns 0.100010 with $\max_t |E_t| = 11.20$, inside the bound. Under Angelopoulos et al.'s tangent integrator the same width returns 0.100001. Raising the saturator's level to $4b$ leaves coverage intact and cuts the $w = 0.999$ excursion from 623.70 to 120.60. That is the first term of $\tau^\star$ moving the edge, not the readout changing behaviour.

What survives of the measurement, and what its status now is. Outside the admissible window the transition is a step. Past $\tau^{\star+}$ the threshold sits permanently below the reachable score range and $|E_t| = (1-\alpha)t$ grows linearly, 60,747 times the bound at $T = 10^6$; the wider bands $\tau = 2.0$, 2.5, 3.0 and 5.0 fail too at $T = 2{\times}10^5$ (Figure 2). **Here** $\sup_x r_t(x) = b$ **and** $\hat{q} \equiv 0$**, so** $\tau^{\star+} = \tau^{\star-} = b/2$, one point of the admissible class, and both terms move it. The legal $\hat{q} \equiv +b/2$ carries the $\tau = 1.5$ band inside the bound against the specified adversary, and against the mirror one no width covers there, the unsmoothed control included. Condition (4) bounds $\sup_x r_t(x)$ only from below, so raising it costs the failure direction nothing; it moves the edge too (Figure 3) and shrinks the excursion, from 623.70 to 20.10 under the tangent integrator Angelopoulos et al. [2023] report using in all their experiments, both at $T = 10^6$. **The method's own integrator therefore sits inside the set at every width run here. The endpoint is where the two boundaries part.** The upper one is closed and the lower one is open, which the null scorecaster cannot show because $\tau^{\star+}$ and $\tau^{\star-}$ coincide there: four saturators with $\tau^{\star+} < \tau^{\star-}$ cover at $\tau = \tau^{\star+}$ exactly under both adversaries, and at $\hat{q} \equiv +0.4$, where $\tau^{\star-} = 0.6 < 1.4 = \tau^{\star+}$, the width 0.599 covers under both while 0.6 fails under the mirror. At the baseline the two coincide at 1, so the primary regime covers there and the mirror adversary does not; Figure 2 and this appendix's primary-regime paragraph print both readings. **The failing arms have** $\mathrm{Sat.} = 1.0000$**.** Condition (4) delivers its full $\pm b$ on every round, and what fails is Proposition 2's other hypothesis: the threshold closing the loop *is* the integrator's output.

Where the rate stops being a tell. At $\tau = 1.9$ with the power-of-two scorecaster the rate is 0.110960 against $\alpha = 0.1$ while $\max_t |E_t|$ is 84,745.70 at $T = 10^6$, so a reader watching only the rate would call that coverage.

Table 3: Legal $(r_t, \hat{q})$ configurations on which the printed $\tau^\star$ over-certifies, each under two legal adversaries. "Proved window" is where (2) certifies retention against every legal adversary; between it and the failure boundaries $\tau^{\star\pm}$ neither direction is claimed, and the rows built on a supremum the dynamics cannot reach or never attain sit there. Here $\emptyset$ marks a configuration for which no width is certified, $\tau = 0$ included. $T = 10^5$ except the two power-of-two rows, at $T = 10^6$. The last row carries no dead band at all.

				$s_t = \tilde{q}_t + \varepsilon$		$s_t \equiv -b/2$	
$(r_t, \hat{q})$ configuration	τ	printed $\tau^\star$	proved window	miscov.	$\max_t \lvert E_t \rvert$	miscov.	$\max_t \lvert E_t \rvert$
$\hat{q} \equiv +0.4$, level b	0.599	1.4	$\tau < 0.6$	0.100010	7.20	0.099890	11.60
$\hat{q} \equiv +0.4$, level b	0.8	1.4	$\tau < 0.6$	0.100050	8.70	**0.000000**	10,000
$\sup_x r_t = 10b$, off the reach	1.001	19	$\tau < 1$	**1.000000**	90,000	0.000000	10,000
$\sup_x r_t$ unbounded, off the reach	1.5	∞	$\tau < 1$	**1.000000**	90,000	0.000000	10,000
$A^+ = 4b$, $A^- = -b$	1.5	7	$\tau < 1$	0.100020	4.30	**0.000000**	10,000
level $4b$, $\hat{q} \equiv 0$	6.999	7	$\tau < 7$	0.100070	12.20	0.100050	11.60
level $4b$, $\hat{q} \equiv 0$	7.0	7	$\tau < 7$	0.100090	12.20	**0.000000**	10,000
$\sup_x r_t = 2b$, unattained	2.99	3	$\tau < 1$	0.100320	211.10	0.098010	210.50
$\sup_x r_t = 2b$, unattained	3.0	3	$\tau < 1$	**1.000000**	90,000	0.000000	10,000
$\sup_x r_t = 2b$, attained	3.0	3	$\tau < 3$	0.100000	12.20	**0.000000**	10,000
$\hat{q}_t = b/2$ on powers of 2	1.0	2	$\emptyset$	0.100009	14.30	**0.000000**	100,000
$\hat{q}_t = b/2$ on powers of 2	1.9	2	$\emptyset$	**0.110960**	84,745.70	0.000000	100,000
$\hat{q} \equiv +b/2$, no band	0	2	$\emptyset$	0.100000	0.90	**0.000000**	10,000

B The four vocabularies

Four vocabularies for one quantity. Four literatures name the movement of a sequentially updated forecast in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC, MQC/SMQC, congruence, nervousness); *jumpiness*, *convergence index*, *(in)consistency*; and *switching cost*. Online conformal prediction gives distribution-free long-run coverage for any bounded scorecaster [Angelopoulos et al., 2023, Thm 1, Prop. 2]. Forecast stability treats the movement as a functional, read out as post-processing [Godahewa et al., 2025, Eq. 3], against a priced decision to switch [Genov et al., 2026, Van Belle et al., 2026, §4.2]. Verification work uses a matched-level design against real producers from 2012 [Pinson and Girard, 2012, Worsnop et al., 2018]. Switching-cost online learning gives worst-case theory for the penalised problem [Andrew et al., 2013, Thm 2].

C Where this sits

Four vocabularies name one quantity, and Appendix B names all four; this appendix places this paper's claim and its concessions among them. **What is conceded, by name.** The placement is prior work: Angelopoulos et al. [2023, Thm. 1, Prop. 2] state coverage, with a rate, for any admissible integrator and any scorecaster, and run a smoothing-family model in that slot themselves; this paper proves that admissible set tight. Dupuy et al. [2026, Appendix A] give the generic argument, and Duerst et al. [2024] adopt that scorecaster. The derivation is prior work too, and this concession matters most. Li et al. [2025, Corollary 2] prove the retention statement in Conformal PID's own notation, with Wang and Hyndman [2024] an independent second instance. Section 2 concedes the smoother object, the metric arithmetic and both readout maps. Zheng et al. [2025], Wu et al. [2025], Chen et al. [2023], Lade et al. [2026] each place a smoother inside their own validity argument. One vocabulary out, this is integrator anti-windup with a feedforward path: free-until-the-bound-binds is its defining specification, not a finding [Galeani et al., 2009]. Nearest in the other direction, Gao et al. [2025] run the same recursion but gate the *model* update on the conformal score. Their one non-standard assumption is that $\sum_{t\le T} q_t \le O(\alpha\, h(T))$ for a sublinear h, supported with a plot rather than derived. Verification's matched-level design tests real producers from 2012, with a control stronger than the pair: the whole marginal [Pinson and Girard, 2012, Worsnop et al., 2018]. Switching-cost online learning adds a Thm. 7 trade-off in which sublinear regret is bought by growing θ with T, which grows the ratio [Andrew et al., 2013, Thm. 7].

The accumulator, read as a bet. Appendix D reads E_t as a game-theoretic capital process and builds a calibration test on top of the boundary above; it is not repeated here.

D The accumulator as a bet, and a calibration test built on the boundary

What is assumed here, and by whom. Everything Section 3 proves is deterministic and pathwise. Proposition 2's window $|E_T| \le c\, h(T) + 1$, and the boundaries (2) that locate its edge, hold on every trajectory a legal adapted adversary can produce; no distributional model on the data appears in either statement, and neither is a statement about a null hypothesis. This section neither uses that nor weakens it. It runs the other way: it takes the error sequence a deployment actually produced as *data*, and tests a hypothesis about it. The hypothesis is imposed here, for the test alone, with $\mathcal{F}_{t-1}$ the history through round $t-1$ (scores, deployed thresholds and errors) against which $\hat{q}_t$ is already required to be predictable:

$$H_0 : \qquad \mathrm{err}_t \mid \mathcal{F}_{t-1} \;\sim\; \mathrm{Bernoulli}(\alpha) \quad \text{for every } t,$$

and it says that the shipped intervals were conditionally calibrated at the target rate. H_0 is the object under test. It is not a premise of Proposition 2, of (2), or of any other claim in this paper; nothing above requires it, implies it, or is disturbed if it fails. When it fails, what loses validity is the test below and nothing else. We are explicit about this because the rest of the paper assumes no distribution at all, and a reader who meets a Bernoulli null on this page must not read it backwards into the pages before it.

The accumulator is a capital process, and it is signed. A Skeptic who each round pays α for the payoff err_t has cumulative net gain exactly E_t, so this paper's accumulator is a game-theoretic capital process at unit stake [Shafer and Vovk, 2019, Ramdas et al., 2023]. It is not an e-value and not an e-process: it is signed, it goes negative on exactly the trajectories the mirror boundary $\tau^{\star-}$ governs, and Proposition 2 bounds it pathwise by feedback rather than under a null by Ville's inequality [Vovk and Wang, 2021]. Nothing below proposes E_t, or a transform of it, as a certificate for any claim this paper makes. What is formal is the standard transform, and it is invoked rather than invented.

The test martingale, and why it is one. Fix λ and set $M_0 = 1$,

$$M_t \;=\; \prod_{i \le t} \big(1 + \lambda(\mathrm{err}_i - \alpha)\big).$$

Each factor takes one of two values, $1 + \lambda(1-\alpha)$ on a miss and $1 - \lambda\alpha$ on a cover, so $M_t \ge 0$ for every path if and only if $\lambda \in [-1/(1-\alpha), 1/\alpha]$, which for $\alpha = 0.1$ is $[-10/9, 10]$. Under H_0, $\mathbb{E}[\mathrm{err}_t \mid \mathcal{F}_{t-1}] = \alpha$, so each factor is conditionally mean one and $\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] = M_{t-1}$: M_t is a nonnegative martingale started at 1, i.e. a test martingale for H_0. Ville's inequality then gives $\mathbb{P}_{H_0}(\sup_t M_t \ge 1/\eta) \le \eta$, so "reject H_0 the first time $M_t \ge 1/\eta$" is an anytime-valid level-η test of the deployed intervals' calibration, at any stopping time and with no horizon fixed in advance. That is a test of the deployment, not of Section 3.

The boundary transfers exactly, because M_t sees only (t, E_t). Since $\mathrm{err}_i \in \{0,1\}$, the product collapses. Writing N_t for the miss count and using $N_t = \alpha t + E_t$,

$$\log M_t \;=\; K(\lambda)\,t \;+\; L(\lambda)\,E_t,$$
$$K(\lambda) = \alpha \log\big(1 + \lambda(1-\alpha)\big) + (1-\alpha)\log(1 - \lambda\alpha),$$
$$L(\lambda) = \log \frac{1 + \lambda(1-\alpha)}{1 - \lambda\alpha}.$$

with $K(\lambda) \le 0$, $K(\lambda) = 0$ only at $\lambda = 0$, and $\operatorname{sign} L(\lambda) = \operatorname{sign} \lambda$. So M_t depends on the path only through the horizon and the accumulator – the same statistic Section 3's dichotomy is stated in – and the transfer is an identity rather than the first-order expansion $\log M_t = \lambda E_t + O(\lambda^2 t)$. It transfers as a dichotomy, not as a measured rate. Inside the retained band $|E_t| \le c\,h(t) + 1$ is logarithmic, so $\log M_t \le K(\lambda)t + |L(\lambda)|(c\,h(t) + 1)$ at every round. Since $K(\lambda) < 0$ the right side falls to $-\infty$ linearly, so $M_T \to 0$ exponentially; and since it is continuous with a single interior maximum it has a finite supremum $S(\lambda)$ over all t, so $\sup_t M_t \le e^{S(\lambda)}$ *uniformly in the horizon*. The test therefore never rejects at any level $\eta < e^{-S(\lambda)}$, however long the deployment runs. It is the uniform bound and not the limit that non-rejection needs, because Ville's inequality prices the supremum. Past either boundary $|E_T|$ is linear in T, so some fixed λ of the sign of E_T makes $\log M_T$ grow linearly and the test rejects at any fixed level within $O(1)$ rounds. Both signs are needed, and that is the point of the two-sided form: the $\tau^{\star+}$ failure drives E_T up and calls for $\lambda > 0$, the mirror failure at $\tau^{\star-}$ drives it down and calls for $\lambda < 0$.

Worked on this paper's own frozen arms, at the null scorecaster. Take the harness of Section 2 ($\alpha = 0.1$, $b = 2$, $c = 1$, $h(t) = \log(t+2)$, $\hat{q} \equiv 0$), where $\tau^{\star+} = \tau^{\star-} = 1$, and the two dead bands that bracket that edge in Table 2 (App. A). *Past the edge*, $\tau = 1.5$ returns miscoverage 1.000000 and $\max_t |E_t| = 900{,}000$ at $T = 10^6$; the run records $n_{\mathrm{err}} = T$ at all four horizons, so the per-round path is exactly $\mathrm{err}_t = 1$ and $E_t = (1-\alpha)t$, and no reconstruction is involved. A genuine pre-registered deployment fixes λ, sign included, before the data arrive – by running both signs as one two-sided test, or by a mixture over λ – and the single sign used per arm below is chosen after the fact, to illustrate the identity on a trajectory already known to fail or to hold; it is not itself a claim of a pre-registered test. At $\lambda = 1$ – a round number well inside $[-10/9, 10]$, not the aggressive endpoint – the identity above gives $M_t = 1.9^{\,t}$, so at level $\eta = 0.05$, threshold $1/\eta = 20$: $M_4 = 13.03$ and $M_5 = 24.76$. **The anytime-valid test rejects on the fifth deployed interval.** At $\eta = 0.001$ it rejects on the eleventh ($M_{10} = 613.11$, $M_{11} = 1164.90$). *Inside the edge*, $\tau = 0.9$ covers at 0.100012 with $\max_t |E_t| = 13.20$ against Proposition 2's 14.8155 at the same $T = 10^6$, and the same $\lambda = 1$ gives $\log M_t \le -0.030639\,t + 9.8632$ from the measured excursion alone: below $\log 20$ from round 225, below 0 from round 322, and $M_T \le e^{-30{,}629}$ at $T = 10^6$. The measured excursion bounds one run; the theorem bounds every one. This width sits inside $\sigma^{\pm}$, so the retention half applies and $|E_t| \le c\,h(t) + 1$ holds at every round, whence $S(1) = \sup_t \big(K(1)t + L(1)(\log(t+2) + 1)\big) = 2.448$, attained at $t = 22$, and $M_t \le 11.57$ at every round and every horizon. The test cannot reject at $\eta = 0.05$ – not merely after round 225, but ever. Five rounds against never, on two arms of one frozen grid. *The mirror direction*, on the same $\tau = 1.5$ configuration read under the strict indicator against $s_t \equiv -b/2$: miscoverage 0.000000, $\mathrm{err}_t = 0$ at every round, $E_t = -\alpha t$, and $\lambda = -1$ gives $M_t = 1.1^{\,t}$, which crosses 20 at $t = 32$ ($M_{31} = 19.19$). The asymmetry is not a defect of the test: a forced miss carries $\log 1.9$ nats per round against a forced cover's $\log 1.1$, because $\alpha = 0.1$ makes misses the rarer event under H_0.

What this adds, and what it does not. It adds a diagnostic and a reading, not a result. The reading is that E_t is the Skeptic's capital and that Section 3 therefore locates, in a quantity a bettor already cares about, the exact width at which a dead band turns a calibrated deployment into one that an anytime-valid test convicts within a handful of rounds. The diagnostic is runnable online by a practitioner who has the errors and not the proof. Neither statement is a claim about E_t's status: E_t is not an e-process, M_t is a textbook transform imposed on top of the result rather than a part of it, and this paper's contribution is the tightness of someone else's admissible radius, proved pathwise. M_t is itself an e-value, and we say so: an e-value for H_0 alone, a hypothesis about one deployment's error sequence, imposed on this page and nowhere else in the paper. No result of this paper is an e-value result or rests on one. Betting enters online conformal inference elsewhere as an optimiser rather than as evidence [Podkopaev et al., 2024].